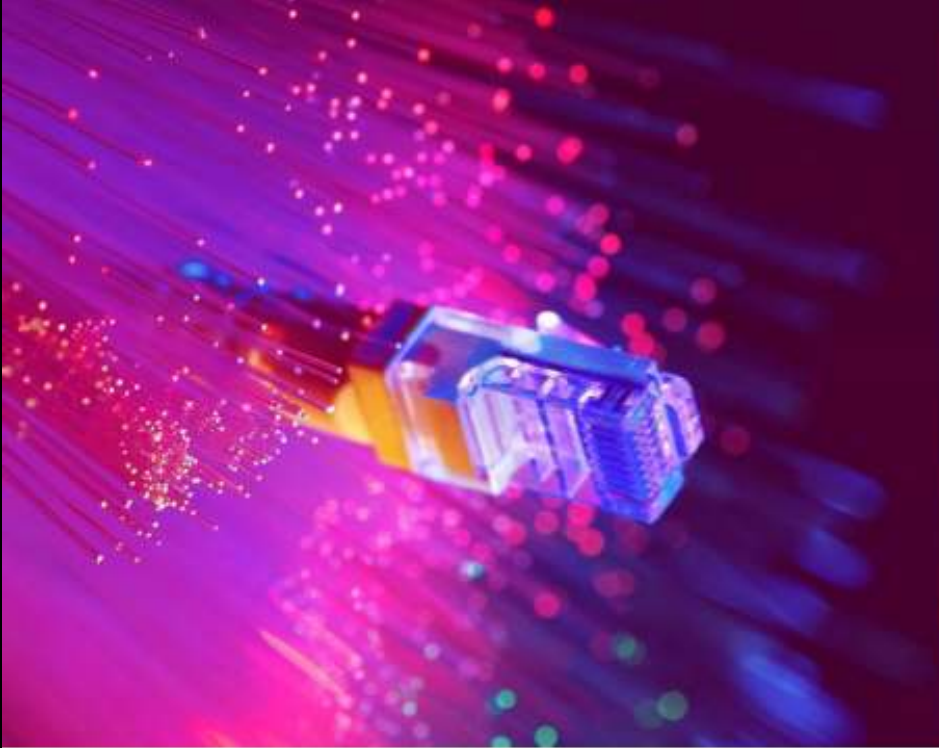




Introduction to ICON climate model

SCC 24

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CSC – Finnish expertise in ICT for research, education and public administration

Introduction to ICON

- Icosahedral Nonhydrostatic Weather and Climate Model
- Next generation modeling system capable of global simulations down to 1km resolutions
- Rewrite of earlier weather model in in the early 2000s
 - 1 600 000 lines of modern Fortran
- Developed my several meteorology institutes
 - DWD, MPI-M, MeteoSwiss, ...
 - ~200 developers
- Operational weather forecasts since 2015

Climate and weather modeling

- Behaviour of the earth system (atmosphere and ocean) can in principle be determined from Navier-Stokes equations driven by the radiation from sun
- Chaotic nature of equations makes weather prediction difficult for more than few days.
- In climate modeling one is interested in statistics, which makes multidecadal simulations feasible.



Simulation by ICON

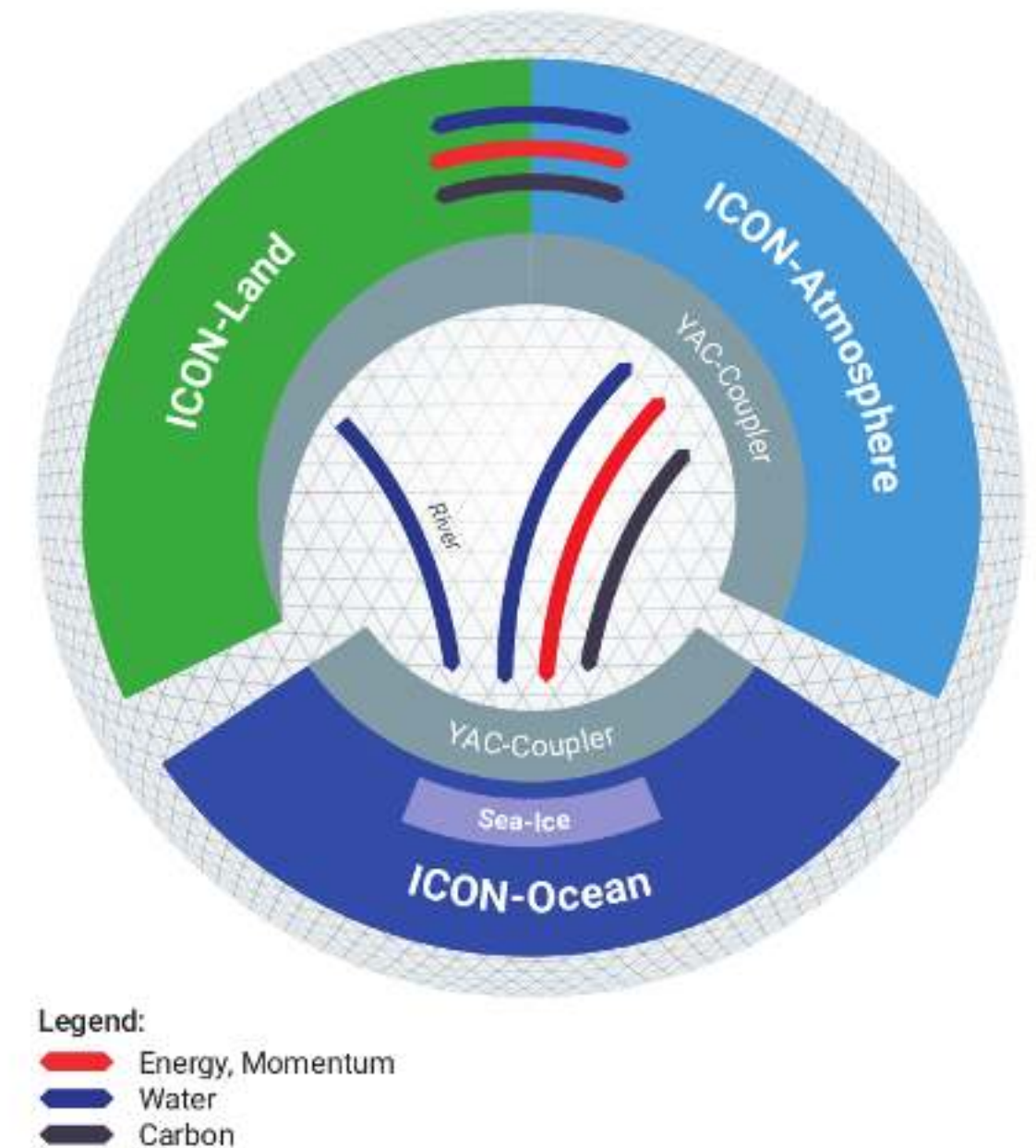
Visualization Niklas Röber, NVIDIA

Climate modeling

- Differential equations are discretized in a grid
- Even though simulation is started from specific initial conditions, climate modeling is fundamentally a boundary value problem
 - same stable climate will develop during the “spin-up” phase.
- In addition to fundamental physical laws, climate and weather models contain parametrizations for physical processes that cannot be presented on the grid resolution
- Prognostic variable: directly integrated from basic equations
- Diagnostic variable: derived from prognostic ones

Model setup

- The most important simulation parameter is the horizontal grid
 - Determines the accuracy of simulation
 - Most variables are also presented in vertical levels
- ICON includes models for atmosphere, land, and ocean, and coupling between them
 - In this exercise we will use only atmosphere and land components.



Grid nomenclature in ICON

- R2B4: 160 km resolution
- R2B5: 80 km resolution
- ...
- R2B11: 1.2 km resolution
- Model performance is often measured as simulated days per day (SDPD)
 - “holy grail” is 365 SDPD or better
- Stable time step is directly proportional to grid resolution, R2B5 requires two times smaller time step than R2B4
 - limits severely weak scaling

Input data in ICON

- Various model parameters
- Initial weather conditions
- Horizontal grid and number of vertical levels
- Ozone content, aerosols, and greenhouse gases in the atmosphere
- Sea surface temperatures and sea ice (especially without ocean)
- Various land related boundary conditions

Building and running ICON

- ICON has some dependencies that may need to be installed separately (eccodes, netcdf, ...)
- Build settings are determined in “config” scripts.
- ICON is normally run via “runscripts”, i.e. shell scripts which contain both the batch job settings and the whole model setup
- ICON can perform I/O asynchronously
 - set of MPI processes is reserved for I/O so that output can be written to disk concurrently to computing
 - Maybe tuning parameter in the competition?